



The magic carpet

A luxury floor for premium ride services - warm, quiet, designed, and it swaps in seconds

Aladdin and the Magic Carpet - Team 4 · Autoneum Challenge 2026 - Final · 2026-09-11



Six in the evening, slush outside: the fare steps into a cabin that is already warm underfoot

Nobody scrubbed it. The mat that earned all day just went back to the depot for a clean one.

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By 2035 the premium robotaxi is a luxury cabin that works sixteen hours a day

USE CASE - THE TARGET

Premium ride services

Uber Black class fleets. The fare pays for a warm, beautiful cabin - most on long journeys.

SEGMENT

C/D-segment BEV, Level 4

Europe and Switzerland. Autoneum's largest market, 46 % of revenue.

AND A BONUS

The private owner

Swaps in a cheap mat when his car works as a taxi. More on him at the end.

Source: Goldman Sachs, Robotaxis to become a \$400 bn market in 2035 (Apr 2026); Autoneum FY2025 revenue release (Jan 2026)



This is what twenty strangers a day leave behind

A private car carries 2-4 people a day. A Waymo does 20+ paid rides - and a premium fare rates what he sees.

Source: Waymo, ~500,000 paid rides a week on ~3,000-3,700 vehicles (TechCrunch, CNBC, Dec 2025 - Mar 2026)

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Cleaning is the one thing a robotaxi cannot do while earning

A BASIC INTERIOR CLEAN

30-60 min

heavy soiling doubles or triples it

PER CLEAN

USD 60-150

routine, up to 500 when bad

FOR THE OPERATOR

= downtime

it earns only while carrying people

Source: Consumer detailing trade estimates, 2025 (not an automotive study); press reporting on Waymo fleet handlers

LEARNING FROM ANOTHER INDUSTRY

A luxury hotel never cleans the mattress

It changes the layer above it. Each layer has one job, and the one that touches the guest is replaced every time.

So the mat comes out at the depot, and the cleaning happens somewhere else

The mat becomes a part the depot swaps - the car never waits for a cleaner.

Wipe-clean skin, mono-material PET

No fibres to hold dirt

Magnetic dock, no visible fasteners

Out in seconds, no tools

Cleaned off the vehicle

The car goes back out, the mat is washed



THE HARDER PROBLEM

But a floor you can wipe is a floor you can hear

Quiet needs an open, porous surface. Clean wants a closed one. Today you pick one.

Autoneum has already solved all of it, just never in the same part

We need	Autoneum has	Where it sits
Washable floor	Washable surface flooring	Trucks (Borgers, 2023)
Wipeable yet acoustic	Alpha-Liner	Wheelhouse, outside
Absorb through a closed skin	RIMIC micro-perforation	Heatshields
Cleanability metric	Cleanability Analyzer	Internal lab
Heating layer	(Heated) Floor mats	Trucks

Source: Autoneum Challenge deck, 7 Sep 2026, slides 6-7; Autoneum product brochure 2021; Autoneum press release on Borgers, Apr 2023

One dock, three tiers - we build the top one, the others fall out of it

Tier 3 is the product for premium fleets. Tiers 1 and 2 are the same dock with less on it.

1 - SHORT DISTANCE

Swap and wipe

Out in seconds, cleaned off the vehicle.

2 - MEDIUM DISTANCE

Quiet and premium

Perforated skin over the absorber. And it looks designed.

3 - LONG DISTANCE

Warm

Heating tracks under the skin. Feel at home.

Luxury already heats its surfaces and sells the mat as a signature

S-Class heated armrests. Rolls-Royce lambswool mats. Never on a swappable floor.

UBER X LUCID X NURO

20,000

luxury-SUV robotaxis from 2026

WAYMO'S BASE CAR

USD 70,000

a Jaguar, since 2018

RIDERS PAY

+13-40 %

Waymo over Uber, 2025-26

Source: Lucid / Uber / Nuro, Jul 2025; Waymo Blog 2018; Obi via TechCrunch, Jan 2026; Mercedes-Benz Warmth Comfort package; Rolls-Royce Boutique

The image shows the interior of a modern, spacious vehicle, possibly a bus or a large van. Four people are seated around a small, round table in the center. From left to right: a woman with dark curly hair, a man with a grey beard, a woman with blonde hair, and a man with dark hair. They are all looking out the large windows at a road that stretches into the distance. The interior has a clean, minimalist design with light-colored walls and ceiling. There are two large skylights on the ceiling. The overall atmosphere is bright and airy.

On a winter journey, warmth is the difference between arriving and charging

Around 200 W of radiant surface instead of 4,500 W of blown air. Winter otherwise costs an EV 20-40 % of its range.

Source: Applied Thermal Engineering 2024; Energy and Buildings 2026; AAA EV range test 2019 (-41 % at -7 C with heating); US DOE VTO 2024

The magic carpet

12/15

SIX IN THE EVENING, AGAIN

Fifty warm cabins a day, and not one of them scrubbed inside the car

The luxury mat runs all week. At the depot it swaps for a clean one in seconds. That is what the fare pays for.

ONE MORE THING

**The same dock lets a private owner
play taxi on a cheap mat and drive
home on his own**

Luxury mat for himself, cheap mat for the shift. Thirty seconds between them.

What we ask the board for

- A pilot: one premium fleet, fifty docks, one winter
- Lab time on the Cleanability Analyzer and an impedance tube
- A decision to treat the dock as a platform, not the mat as a part

Sources: Autoneum challenge deck 2026, product brochure 2021, FY2025 release; Goldman Sachs 2026; Obi / TechCrunch 2026; Lucid-Uber-Nuro 2025; AAA 2019, US DOE 2024; Applied Thermal Engineering 2024, Energy and Buildings 2026. Full references in the presenter script.